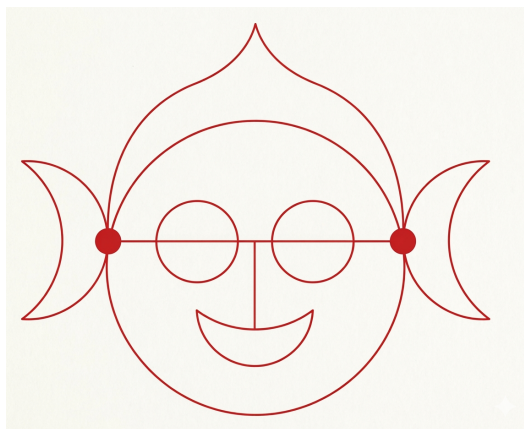


Combinatorics Problem Set

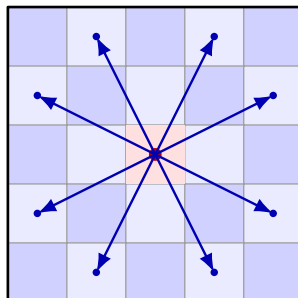
BdMO National Camp 2026

1. How many board positions can appear at the end of a valid classical 3×3 tic-tac-toe game? Treat all symmetric positions as distinct.
2. How many valid tic-tac-toe games can be played on a 3×3 grid? Treat all symmetric game histories as distinct, and stop counting exactly when a win or draw occurs.
3. Triangle counting in a modified tetrakis-style tiling / SKY puzzle configuration. Count the total number of triangles using the following methods:
 - (a) induction / counting in the SKY puzzle style,
 - (b) counting using symmetry,
 - (c) adjacency-based counting.
4. Fibonacci: history and musical bit. Determine the number of tilings in each case:
 - (a) a $1 \times n$ grid using 1×1 and 1×2 blocks,
 - (b) a 1×18 grid using 1×2 and 1×3 blocks,
 - (c) a 1×10 grid using 1×1 and 1×3 blocks,
 - (d) a 1×20 grid using 1×2 blocks.
5. In a tic-tac-toe game, the first move of each player (that is, the first pair of moves) is chosen uniformly at random. After that, both players play optimally. What is the probability that the first player wins?
6. In how many ways can the following diagram be drawn without lifting the pen and without overwriting or retracing any line? Use Eulerian-path ideas.



7. How many ways are there to fill a 4×4 grid, leaving one square blank, with the numbers 1 to 15 so that whenever two squares contain consecutive integers, those two squares are a knight's move apart?

The knight moves allowed are shown below.



8. A 3×3 tic-tac-toe game is played with the extra rule that no player may move in the center square on the first move. Under optimal play, determine the result: win, lose, or draw.
9. Tic-tac-toe is played using the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9 instead of X and O . A player wins by selecting three numbers in a line whose sum is 15. Analyze the game under each of the following rules:
- normal play,
 - 5 is forbidden for Player 1 on the first move, but Player 2 may use 5 on the first move.
10. Show that there exist exactly 880 distinct normal 4×4 magic squares when all symmetric cases are counted as distinct.